UWIRS x ESS Robotics Design Competition Rubric

Assessment Rubric (Analytic Marking Scheme)

Stage 1 — Proposal Stage (Formative – Feedback Only)

Criterion	Sub-Criteria	Marks	Full Marks Standard
Problem Definition & Context	Clear problem statement & stakeholders	10	Problem precisely defined with stakeholders identified
	Engineering constraints & assumptions	10	Constraints realistic and justified
	Measurable goals & outcomes	10	Quantified success metrics stated
Subtotal		**30**	
Conceptual Robotics Approach	Robot/platform selection & justification	15	Choice technically appropriate and justified
	High-level architecture diagram	15	Clear subsystem blocks and interactions
	Feasibility reasoning	10	Demonstrates technical plausibility
Subtotal		**40**	
Sustainability Alignment	Clear SDG linkage	10	Direct connection to SDG problem
	Expected impact reasoning	10	Explains how solution improves sustainability
Subtotal		**20**	
Communication & Structure	Organization & clarity	5	Logical, easy to follow
	Diagrams & professionalism	5	Clean visuals and formatting
Subtotal		**10**	

Stage 2 — Final Design Stage (Graded – 100%)

A. Robotics Engineering Competency (80)

Criterion	Weight	Sub-Criteria	Marks	Full Marks Standard
Problem Statement & Requirements	**10%**	Problem context & motivation	3	Real-world need clearly established
		Engineering requirements (quantified)	4	Measurable targets defined
		Constraints & justified assumptions	3	Practical limits identified and defended
System Architecture & Design Methodology	**20%**	System block diagrams & signal/data flow	6	Complete labelled diagrams with all subsystems
		Subsystem decomposition & interfaces	7	Clear modular breakdown with defined roles
		Design workflow & decision justification	7	Logical engineering process with trade-offs explained
Robotics Technical Depth	**25%**	Sensors selection & justification	6	Appropriate sensors with technical reasoning/specifications
		Actuators/mechanics design	6	Correct actuation method with performance justification
		Control strategy/algorithms	6	Clear control logic/flowcharts/math explanation
		Power & energy system considerations	4	Battery/consumption analysis included

		Safety & reliability considerations	3	Hazards and protections addressed
Modelling, Analysis & Validation	**15%**	Mathematical/engineering calculations	5	Correct, relevant quantitative analysis
		Simulation or performance modelling	5	Simulations or structured evaluation provided
		Results interpretation & comparison	5	Insights drawn, alternatives compared
Integration & Feasibility	**10%**	Subsystem integration reasoning	4	Clear interconnections and compatibility
		Practical implementation plan	3	Realistic deployment strategy
		Risks, limitations & mitigation	3	Risks identified with solutions

B. Sustainability & SDG Impact (20)

Criterion	Weight	Sub-Criteria	Marks	Full Marks Standard
SDG Relevance & Problem Fit	**8%**	Clear SDG identification	3	Correctly chosen and justified
		Alignment with SDG targets	5	Directly addresses core issue
Environmental & Social Impact	**7%**	Environmental impact analysis	3	Energy/resource effects discussed
		Social/community benefit	2	Clear user benefit
		Quantification of impact	2	Measurable improvement estimated
Ethical & Responsible Engineering	**5%**	Safety & user protection	2	Hazards minimized
		Accessibility/affordability	2	Inclusive and practical design
		Maintainability/lifecycle	1	Long-term usability considered

Final Mark Summary

Section	Marks
Robotics Engineering Competency	80
Sustainability & SDG Impact	20
TOTAL	**100**